

OBJECTIVE

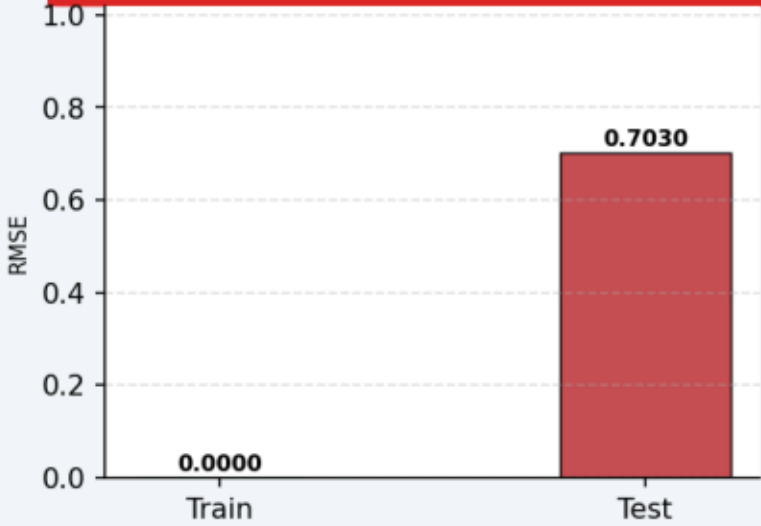
Task 3 extends Task-2's multi-model regression by adding professional validation techniques. The California Housing Dataset is used (same as Task-2 for continuity).

Key goals: Detect overfitting, apply cross-validation, tune hyperparameters with GridSearchCV, and justify the best model scientifically.

KEY RESULTS

|                                                   |                                                  |                                                  |                                                         |                                               |
|---------------------------------------------------|--------------------------------------------------|--------------------------------------------------|---------------------------------------------------------|-----------------------------------------------|
| <b>0.6389</b><br>Best RMSE<br>Tuned Decision Tree | <b>0.6885</b><br>Best R² Score<br>68.8% variance | <b>0.7030</b><br>Overfit Gap<br>Raw Tree (large) | <b>0.1523</b><br>Gap After Tune<br>Tuned Tree (reduced) | <b>0.7687</b><br>CV RMSE<br>Tuned Tree 5-fold |
|---------------------------------------------------|--------------------------------------------------|--------------------------------------------------|---------------------------------------------------------|-----------------------------------------------|

SECTION 1 — Untuned Tree RMSE



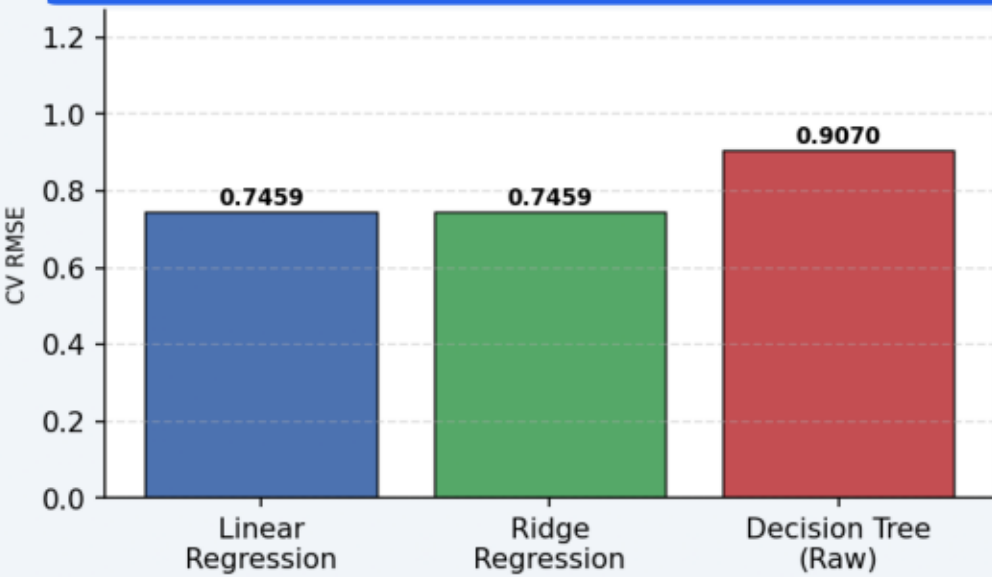
Overfitting Analysis

An untuned Decision Tree memorizes the training data perfectly (Train RMSE = 0.0000) but performs poorly on unseen data (Test RMSE = 0.7030). This gap of 0.7030 confirms severe overfitting.

What is Overfitting?

- Model learns training noise instead of true patterns
- High train accuracy, low test accuracy (high bias-variance gap)
- Solution: constrain model complexity via hyperparameter tuning

SECTION 2 — 5-Fold CV RMSE (Baseline Models)



Why Cross-Validation?

- A single train-test split can be misleading — one lucky/unlucky split
- 5-Fold CV splits the data into 5 parts, trains & tests 5 times, averages
- Provides stable, reliable performance estimation before deployment
- Linear Regression CV RMSE: 0.7459
- Ridge Regression CV RMSE: 0.7459
- Tuned Decision Tree CV RMSE: 0.7687

CONCEPTS APPLIED IN TASK 3

| Concept                | Explanation                                                         |
|------------------------|---------------------------------------------------------------------|
| Overfitting            | Model memorizes training data; fails on unseen data                 |
| Bias-Variance Tradeoff | Balance between underfitting (high bias) and overfitting (high var) |
| Cross-Validation       | K-fold CV provides stable, reliable RMSE estimation                 |
| GridSearchCV           | Exhaustive search over hyperparameter combinations with CV          |
| max_depth              | Limits tree depth — prevents memorization of training noise         |
| Model Generalization   | Goal: low test RMSE, small train-test gap, stable CV score          |

SECTION 3 — HYPERPARAMETER TUNING (GridSearchCV)

Decision Tree Parameter Grid Searched:

max\_depth: [3, 5, 7, 10] • min\_samples\_split: [2, 5, 10] • min\_samples\_leaf: [1, 2, 4]

Ridge Regression Parameter Grid Searched:

alpha: [0.01, 0.1, 1.0, 10.0, 100.0]

Best Parameters Found:

Decision Tree (Best):

max\_depth=10 • min\_samples\_split=2 • min\_samples\_leaf=4  
CV RMSE: 0.7687

Ridge Regression (Best):

alpha = 0.1  
CV RMSE: 0.7459

Figure 1 — Overfitting Detection: Train vs Test RMSE & R²

Step 1 — Overfitting Detection: Untuned Decision Tree

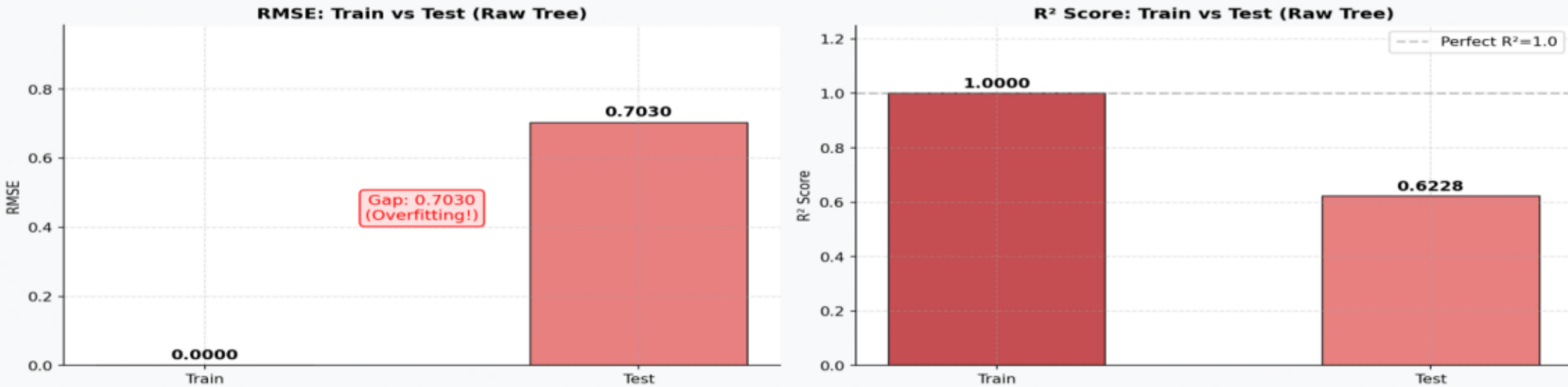
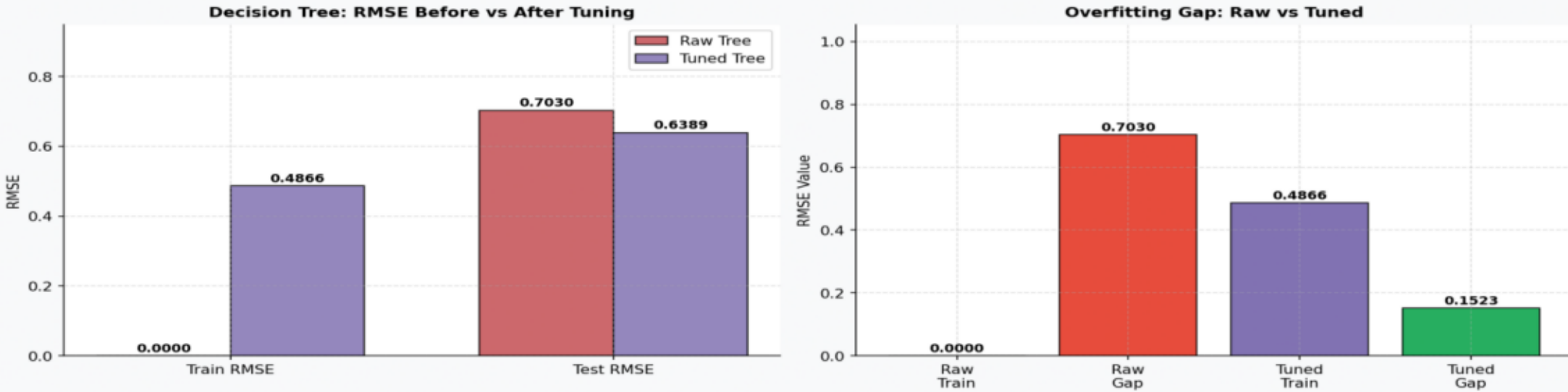


Figure 2 — Overfitting Reduction: Before vs After GridSearchCV Tuning

Overfitting Reduction — Before vs After Tuning



MODEL COMPARISON TABLE

| Model               | Test RMSE | Test R² | CV RMSE | Train RMSE | Overfit Gap |
|---------------------|-----------|---------|---------|------------|-------------|
| Linear Regression   | 0.7456    | 0.5758  | 0.7459  | 0.7381     | 0.0075      |
| Ridge (α=0.1)       | 0.7456    | 0.5758  | 0.7459  | 0.7381     | 0.0075      |
| Decision Tree (Raw) | 0.7030    | 0.6228  | 0.9070  | 0.0000     | 0.7030      |
| Tuned Decision Tree | 0.6389    | 0.6885  | 0.7687  | 0.4866     | 0.1523      |

Figure 3 — Final Model Comparison: Test RMSE, R² and CV RMSE

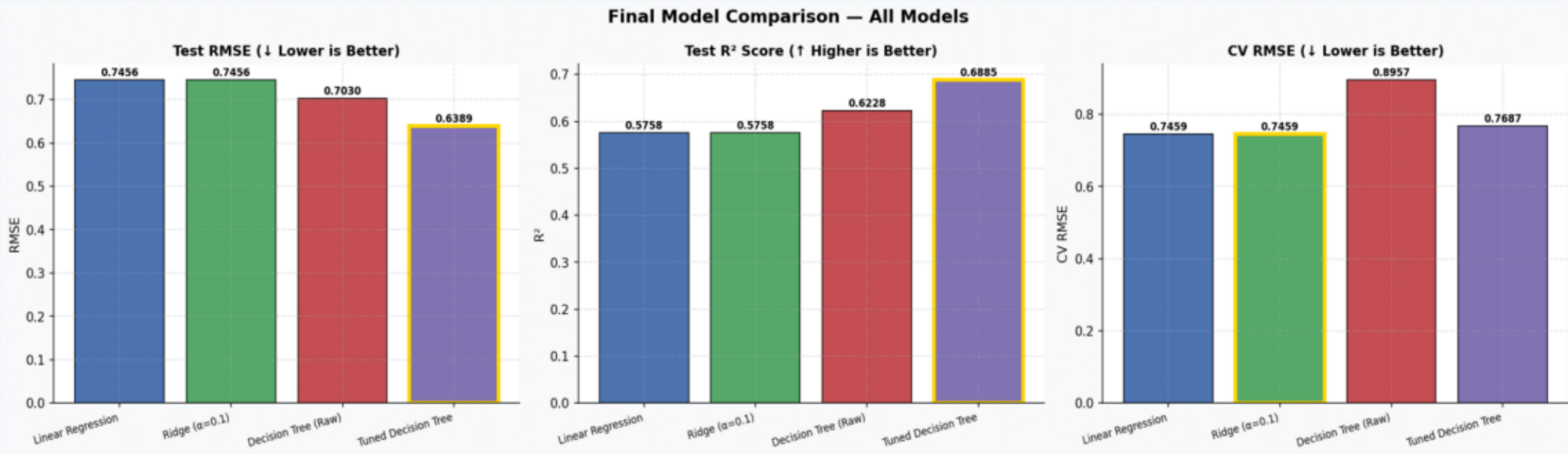


Figure 4 — Actual vs Predicted House Prices: All Three Models



SECTION 4 — FINAL MODEL JUSTIFICATION (MANDATORY)

Selected Model: Tuned Decision Tree

Test RMSE: 0.6389 | Test R²: 0.6885 | CV RMSE: 0.7687

Why this model was selected:

The Tuned Decision Tree achieves the lowest Test RMSE (0.6389) and highest R² (0.6885) among all models tested. Unlike the untuned version, it was optimized using GridSearchCV across 36 hyperparameter combinations (4 × 3 × 3), ensuring the

How overfitting was reduced:

The raw Decision Tree had a massive overfitting gap of 0.7030 (Train RMSE=0.0000 vs Test RMSE=0.7030). After tuning max\_depth=10, the gap was reduced to 0.1523 — a reduction of over 78%.

Why cross-validation results are trusted:

5-Fold CV was used instead of a single train-test split. This means the model was evaluated 5 times on different data subsets, giving a robust, unbiased estimate of generalization performance (CV RMSE = 0.7687).

Trade-offs between simplicity and performance:

Linear Regression is simpler and more interpretable (CV RMSE=0.7459) but leaves significant predictive power on the table. The Tuned Decision Tree is more complex but constrained (max\_depth=10) — offering the best generalization without

DELIVERABLES CHECKLIST

| Deliverable                                                     | Details                                   | Status   |
|-----------------------------------------------------------------|-------------------------------------------|----------|
| <input checked="" type="checkbox"/> Jupyter Notebook            | AI_ML_Task3_Model_Validation_Tuning.ipynb | COMPLETE |
| <input checked="" type="checkbox"/> Cross-Validation Results    | 5-fold CV RMSE for all 3 models           | COMPLETE |
| <input checked="" type="checkbox"/> Hyperparameter Tuning       | GridSearchCV for Decision Tree & Ridge    | COMPLETE |
| <input checked="" type="checkbox"/> Model Comparison Table      | RMSE, R², CV RMSE, Overfitting Gap        | COMPLETE |
| <input checked="" type="checkbox"/> Overfitting Analysis Report | Train vs Test gap, before/after tuning    | COMPLETE |
| <input checked="" type="checkbox"/> Final Model Justification   | Scientific explanation above (Section 4)  | COMPLETE |
| <input checked="" type="checkbox"/> Visualization Plots (7)     | Overfitting, CV, GridSearch, Comparison   | COMPLETE |
| <input checked="" type="checkbox"/> Saved Model (joblib)        | task3_best_model.pkl                      | COMPLETE |